test

September 8, 2026

[1]:
```
from sentence_transformers import SentenceTransformer
from transformers import AutoModel
import sentence_transformers
import transformers


print(f"sentence_transformers version: {sentence_transformers.__version__}")
print(f"transformers version: {transformers.__version__}")
```

```
/root/kyx/contextual-embeddings/.venv/lib/python3.12/site-
packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update
jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm

sentence_transformers version: 5.3.0
transformers version: 5.3.0
```

[2]:
```
sentences = ["", "This is an example sentence", "Each sentence is converted"]

model = SentenceTransformer('nishimoto/contriever-sentencetransformer')
embeddings = model.encode(sentences)
print(embeddings)
print(model)
print(f"#params: {sum(p.numel() for p in model.parameters()) / 1e6:.2f} M")
```

```
Loading weights: 100%|      | 199/199 [00:00<00:00, 26349.29it/s]

[[-0.05403396  0.0072601  -0.03522868 …  0.00571051 -0.03750556
  -0.02870593]
 [-0.02897058  0.00683418 -0.04599153 …  0.00435537  0.06911174
  -0.03234531]
 [ 0.0376888  -0.07167679 -0.01861148 … -0.01329108  0.01142806
  -0.03365104]]
SentenceTransformer(
  (0): Transformer({'max_seq_length': 509, 'do_lower_case': False,
'architecture': 'BertModel'})
  (1): Pooling({'word_embedding_dimension': 768, 'pooling_mode_cls_token':
False, 'pooling_mode_mean_tokens': True, 'pooling_mode_max_tokens': False,
'pooling_mode_mean_sqrt_len_tokens': False, 'pooling_mode_weightedmean_tokens':
```

```
False, 'pooling_mode_lasttoken': False, 'include_prompt': True})
)
#params: 109.48 M
```

[3]:
```
sentences = ["","This is an example sentence", "Each sentence is converted"]

model = SentenceTransformer('lightonai/modernbert-embed-large')
embeddings = model.encode(sentences)
print(embeddings)
print(model)
print(f"#params: {sum(p.numel() for p in model.parameters()) / 1e6:.2f} M")
```

```
Loading weights: 100%|      | 170/170 [00:00<00:00, 4317.40it/s]
[[ 3.2551438e-02 -7.8945326e-05  3.7624785e-03 …  3.1581555e-02
   1.4211968e-02 -2.4172228e-02]
 [ 4.8267553e-03  1.5978618e-02  1.0223079e-02 …  2.1176031e-02
   1.2241008e-02  4.0973876e-02]
 [ 3.6105409e-02  9.3649756e-03  6.0103661e-03 …  2.0348933e-02
   2.6452907e-03  3.2300778e-02]]
SentenceTransformer(
  (0): Transformer({'max_seq_length': 8192, 'do_lower_case': False,
'architecture': 'ModernBertModel'})
  (1): Pooling({'word_embedding_dimension': 1024, 'pooling_mode_cls_token':
False, 'pooling_mode_mean_tokens': True, 'pooling_mode_max_tokens': False,
'pooling_mode_mean_sqrt_len_tokens': False, 'pooling_mode_weightedmean_tokens':
False, 'pooling_mode_lasttoken': False, 'include_prompt': True})
  (2): Normalize()
)
#params: 394.78 M
```

[4]:
```
model_ctx = AutoModel.from_pretrained(
    "perplexity-ai/pplx-embed-context-v1-0.6B",
    trust_remote_code=True
)

doc_chunks = [
    [
        "Curiosity begins in childhood with endless questions about the world.",
        "As we grow, curiosity drives us to explore new ideas.",
        "Scientific breakthroughs often start with a curious question."
    ],
    [
        "The curiosity rover explores Mars searching for ancient life.",
        "Each discovery on Mars sparks new questions about the universe."
    ]
]
# Returns list of numpy arrays (one per document)
```

```python
# embeddings[0].shape = (3, 1024), embeddings[1].shape = (2, 1024)
embeddings = model_ctx.encode(doc_chunks)

print(embeddings)
print(model_ctx)
print(f"#params: {sum(p.numel() for p in model_ctx.parameters()) / 1e6:.2f} M")
```

```
You are using a model of type `bidirectional_pplx_qwen3` to instantiate a model
of type ``. This may be expected if you are loading a checkpoint that shares a
subset of the architecture (e.g., loading a `sam2_video` checkpoint into
`Sam2Model`), but is otherwise not supported and can yield errors. Please verify
that the checkpoint is compatible with the model you are instantiating.
Loading weights: 100%|      | 310/310 [00:00<00:00, 4618.75it/s]
```

```
[array([[-26.       ,  -9.       ,   0.       , …,   3.       ,
         -3.       , -26.       ],
       [-13.       ,  -2.       ,   0.       , …,  14.       ,
         -4.       , -13.       ],
       [ -2.       , -14.       ,   0.       , …,   9.       ,
          0.       ,  -1.9999999]], dtype=float32), array([[ 24., -32.,   0.,
…,  -4., -21.,  -6.],
       [ 13.,  10.,   0., …,   8., -15., -11.]], dtype=float32)]
PPLXQwen3ContextualModel(
  (embed_tokens): Embedding(151936, 1024)
  (layers): ModuleList(
    (0-27): 28 x Qwen3DecoderLayer(
      (self_attn): Qwen3Attention(
        (q_proj): Linear(in_features=1024, out_features=2048, bias=False)
        (k_proj): Linear(in_features=1024, out_features=1024, bias=False)
        (v_proj): Linear(in_features=1024, out_features=1024, bias=False)
        (o_proj): Linear(in_features=2048, out_features=1024, bias=False)
        (q_norm): Qwen3RMSNorm((128,), eps=1e-06)
        (k_norm): Qwen3RMSNorm((128,), eps=1e-06)
      )
      (mlp): Qwen3MLP(
        (gate_proj): Linear(in_features=1024, out_features=3072, bias=False)
        (up_proj): Linear(in_features=1024, out_features=3072, bias=False)
        (down_proj): Linear(in_features=3072, out_features=1024, bias=False)
        (act_fn): SiLUActivation()
      )
      (input_layernorm): Qwen3RMSNorm((1024,), eps=1e-06)
      (post_attention_layernorm): Qwen3RMSNorm((1024,), eps=1e-06)
    )
  )
  (norm): Qwen3RMSNorm((1024,), eps=1e-06)
  (rotary_emb): Qwen3RotaryEmbedding()
  (_flexible_quantizer): FlexibleQuantizer(
    (_int8_quantizer): Int8TanhQuantizer()
```

```
    (_binary_quantizer): BinaryTanhQuantizer()
  )
)
#params: 596.05 M
```

[ ]:
```python
model = SentenceTransformer(
    "perplexity-ai/pplx-embed-v1-0.6B",
    trust_remote_code=True
)

texts = [
    "",
    "Scientists explore the universe driven by curiosity.",
    "Children learn through curious exploration.",
    "Historical discoveries began with curious questions.",
    "Animals use curiosity to adapt and survive.",
    "Philosophy examines the nature of curiosity.",
]

embeddings = model.encode(texts) # Shape: (5, 1024), quantized to int8
print("embeddings (int8):", embeddings)

embeddings = model.encode(texts, quantization="binary") # Shape: (5, 1024),␣
 ↪quantized to binary
print("embeddings (binary):", embeddings)

print(model)
print(f"#params: {sum(p.numel() for p in model.parameters()) / 1e6:.2f} M")
```

```
Loading weights: 100%|      | 310/310 [00:00<00:00, 5918.52it/s]
embeddings (int8): [[  0.   0.   0. …   0.   0.   0.]
 [-21. -31.   0. …  30.  -3. -21.]
 [-37.  27.   0. …  22.   5. -15.]
 [ -8.  -6.   0. …   5.  11. -33.]
 [-51.  26.   0. …  46. -14. -14.]
 [  9. -18.   0. …  57.  13.  -9.]]
embeddings (binary): [[ 1.          1.          1.         …  1.          1.
   1.        ]
 [-1.         -1.         -0.99999994 …  1.         -1.
  -1.        ]
 [-1.          1.         -1.         …  1.          1.
  -0.99999994]
 [-0.99999994 -0.99999994 -1.         …  1.          1.
  -1.        ]
 [-1.          0.99999994 -1.         …  1.         -1.
  -1.        ]
 [ 1.         -0.99999994 -1.         …  1.          1.
```

```
  -0.99999994]]
SentenceTransformer(
  (0): Transformer({'max_seq_length': 32768, 'do_lower_case': False,
'architecture': 'PPLXQwen3Model'})
  (1): Pooling({'word_embedding_dimension': 1024, 'pooling_mode_cls_token':
False, 'pooling_mode_mean_tokens': True, 'pooling_mode_max_tokens': False,
'pooling_mode_mean_sqrt_len_tokens': False, 'pooling_mode_weightedmean_tokens':
False, 'pooling_mode_lasttoken': False, 'include_prompt': True})
  (2): FlexibleQuantizer(
    (_int8_quantizer): Int8TanhQuantizer()
    (_binary_quantizer): BinaryTanhQuantizer()
  )
)
#params: 596.05 M
```

The Kernel crashed while executing code in the current cell or a previous cell.

Please review the code in the cell(s) to identify a possible cause of the␣ ↪failure.

Click here for more info.

View Jupyter log for further details.